Lecture Notes: Measurement, Matter and Mathematics

Unit 1: Measurement and Units

A measurement is only useful when its unit is clear. In this unit we meet the units that carry powers and the habits that keep calculations honest.

Quantities and Their Units

Many physical quantities are ratios of other quantities, and their units show it. Reading a unit from left to right tells how the quantity is built.

Definition. A unit is a standard quantity used to express the size of a measurement.

Powers in Compound Units

The plate has an area of 2.0 m^2 and a thickness of 4 cm, so its volume is 0.08 m^3. The solution contains 0.1 mol dm^{-3} of glucose, so 250 cm^3 of it holds 0.025 mol. The reservoir holds 11.0 km^3 of water spread over 40 km^2. A cart accelerates at 4.5 m/s^2 for 6 s and reaches a speed of 27 m/s.

A steel wire with a cross-section of 10.0 mm^2 carries a load of 900 N, which is a stress of 90 N/mm^2. The specific heat capacity of water is 4.18 J g^{-1} K^{-1}. The pump delivers 2.2 m^3/s, which is 2200 dm^3 every second. At sea level the density of air is about 1.2 kg m^{-3}, and a barometer reading corresponds to a pressure of 1.2 times 10^5 N/m^2.

The density of iron is 7.87 g/cm^3, so a block of volume 60 cm^3 has a mass of 472 g. The intensity of sunlight at the ground is roughly 950 W m^{-2} on a clear day. The measured viscosity of the aluminium melt is 4.0 kg m^{-1} s^{-1}. Near the surface of the earth, free fall has an acceleration of 9.8 m s^{-2}.

The specific heat capacity of copper is 0.39 J g^{-1} K^{-1}.

Why does the unit matter?

Example

A cart rolls 12 m down a ramp in 4.0 s, starting from rest. Find its acceleration.

Solution

Use $d = (1/2)\,a\,t^2$, so $a = 2d / t^2 = 2(12) / (4.0)^2 = 1.5$ m/s^2.

Density, Flow and Intensity

The intensity of sunlight at the ground is roughly 340 W m^{-2} on a clear day. The measured viscosity of the granite melt is 0.5 kg m^{-1} s^{-1}. Near the surface of the earth, free fall has an acceleration of 9.81 m s^{-2}. A steel wire with a cross-section of 2.0 mm^2 carries a load of 1500 N, which is a stress of 750 N/mm^2.

At sea level the density of air is about 1.2 kg m^{-3}, and a barometer reading corresponds to a pressure of 1.3 times 10^5 N/m^2. A cart accelerates at 3.5 m/s^2 for 4 s and reaches a speed of 14 m/s. The plate has an area of 2.0 m^2 and a thickness of 5 cm, so its volume is 0.1 m^3. The density of iron is 7.87 g/cm^3, so a block of volume 40 cm^3 has a mass of 315 g.

The reservoir holds 2.5 km^3 of water spread over 85 km^2. The pump delivers 0.4 m^3/s, which is 400 dm^3 every second. The solution contains 2.0 mol dm^{-3} of glucose, so 250 cm^3 of it holds 0.5 mol. A cart accelerates at 2.5 m/s^2 for 8 s and reaches a speed of 20 m/s.

The density of brass is 8.50 g/cm^3, so a block of volume 40 cm^3 has a mass of 340 g.

Remember: the exponent applies to the whole unit.

Always convert before you compute.

Convert every quantity into one system of units before any number is substituted.

Key Takeaways

- A rate of change is a ratio of two differences.
- Neighbouring table rows give the best estimate of a local rate.
- State the unit of every rate you report.

Unit 2: Matter and Reactions

Matter is made of atoms, and atoms combine in fixed proportions. Formulas record these proportions and the electric charge that ions carry.

Atoms, Molecules and Ions

Definition. An ion is an atom or group of atoms that carries a net electric charge.

Sodium bicarbonate has the formula $NaHCO_3$ and is stored in a sealed bottle. In the reaction NH_3 reacts with $Al_2(SO_4)_3$ to give $Ca(OH)_2$, the mass of the products equals the mass of the reactants. Glucose, $C_6H_{12}O_6$, is made of molecules or ions that differ from those of aluminium sulfate, $Al_2(SO_4)_3$.

When the salt of sodium and chloride dissolves in water it releases Na^+ and Cl^- ions. The copper(II) ion, Cu^{2+}, is formed when the neutral species loses 2 electrons. A bottle labelled chloride holds the ion Cl^-, which is surrounded by water molecules.

A solution of water (H_2O) was mixed with a solution containing Ca^{2+} ions. When the salt of iron(III) and chloride dissolves in water it releases Fe^{3+} and Cl^- ions. A bottle labelled sulfate holds the ion SO_4^{2-}, which is surrounded by water molecules.

Ammonia, NH_3, is made of molecules or ions that differ from those of methane, CH_4.

Charges on Ions

The calcium ion, Ca^{2+}, is formed when the neutral species loses 2 electrons. A solution of sulfuric acid (H_2SO_4) was mixed with a solution containing Na^+ ions. In the reaction H_2O reacts with H_2SO_4 to give $C_6H_{12}O_6$, the mass of the products equals the mass of the reactants.

Aluminium sulfate has the formula $Al_2(SO_4)_3$ and is stored in a sealed bottle. In the reaction C_2H_6O reacts with $Al_2(SO_4)_3$ to give $Ca(OH)_2$, the mass of the products equals the mass of the reactants. A bottle labelled sulfate holds the ion SO_4^{2-}, which is surrounded by water molecules.

Sulfuric acid has the formula H_2SO_4 and is stored in a sealed bottle. Glucose, $C_6H_{12}O_6$, is made of molecules or ions that differ from those of aluminium sulfate, $Al_2(SO_4)_3$. A solution of glucose ($C_6H_{12}O_6$) was mixed with a solution containing Na^+ ions.

The iron(III) ion, Fe^{3+}, is formed when the neutral species loses 3 electrons.

Is the charge balanced?

Solutions and Reactions

Reactions rearrange atoms without destroying them. Balanced equations keep count of every atom and every charge.

Example

A solution is made by dissolving 18 g of salt in water to make 250 cm^3. Find the concentration in g per cm^3.

Solution

The concentration is 18 / 250 = 0.072 g/cm^3.

Key Takeaways

- Area under a velocity graph is displacement.
- Regions below the axis count as negative.
- Check the answer against a rough area estimate.

Unit 3: Nuclei and Radioactivity

Nuclei are described by two integers, one for the protons and one for the protons and neutrons together. Both numbers are written before the symbol.

Isotopes and Nuclear Notation

Definition. An isotope is a form of an element with a particular number of neutrons.

The nucleus ^{238}U decays by alpha emission, and its half-life $t_{1/2}$ is 4.5 billion years. Radium-226, written ^{226}Ra, contains 88 protons and 138 neutrons. The isotopes ^{131}I and ^{132}I of iodine have the same chemical behaviour but different masses. A sample of ^{32}P loses half of its atoms every 14.3 days, so the activity falls in a regular pattern.

The nucleus ^{241}Am decays by alpha emission, and its half-life $t_{1/2}$ is 432 years. Strontium-90, written ^{90}Sr, contains 38 protons and 52 neutrons. The isotopes ^{222}Rn and ^{221}Rn of radon have the same chemical

behaviour but different masses. A sample of ^{14}C loses half of its atoms every 5730 years, so the activity falls in a regular pattern.

The nucleus ^{137}Cs decays by beta emission, and its half-life $t_{1/2}$ is 30.2 years. Polonium-210, written ^{210}Po, contains 84 protons and 126 neutrons.

Check that mass number and atomic number are conserved:

Decay and Half-Life

Radioactive nuclei decay at random, yet a large sample decays in a predictable pattern. The half-life is the usual measure of that pattern.

The isotopes ^{3}H and ^{5}H of hydrogen have the same chemical behaviour but different masses. A sample of ^{60}Co loses half of its atoms every 5.27 years, so the activity falls in a regular pattern. The nucleus ^{238}U decays by alpha emission, and its half-life $t_{1/2}$ is 4.5 billion years. Radium-226, written ^{226}Ra, contains 88 protons and 138 neutrons.

The isotopes ^{131}I and ^{132}I of iodine have the same chemical behaviour but different masses. A sample of ^{32}P loses half of its atoms every 14.3 days, so the activity falls in a regular pattern. The nucleus ^{241}Am decays by alpha emission, and its half-life $t_{1/2}$ is 432 years. Strontium-90, written ^{90}Sr, contains 38 protons and 52 neutrons.

The isotopes ^{222}Rn and ^{221}Rn of radon have the same chemical behaviour but different masses. A sample of ^{14}C loses half of its atoms every 5730 years, so the activity falls in a regular pattern.

Example

A data set has the values 4.2, 5.1 and 6.0. Find the mean.

Solution

The mean is (4.2 + 5.1 + 6.0) / 3 = 5.1.

Unit 4: Sequences and Statistics

A sequence is a list with an order, and the index of a term says where it stands. Subscripts and superscripts carry much of the notation.

Sequences and Series

Definition. A sequence is an ordered list of numbers given by a rule.

The null hypothesis H_0 is rejected when the p-value is below 0.05. Taking logarithms to base ten, $\log_{10} 450$ is about 2.65. The point (x_1, y_1) lies on the line, and so does (x_2, y_2). The coefficient of determination R^2 equals 0.97 for this fit.

The 1st, 2nd and 3rd terms are 9, 5 and 7. Expanding $(a + b)^2$ gives $a^2 + 2ab + b^2$. The kinetic energy E_k of the cart is 27 J at the bottom of the ramp. A tolerance of 10^{-3} is enough for the 17th iteration of the method.

Each term satisfies $a_{n+1} = a_n + 3$, so with $a_1 = 7$ the 5th term is 19. The half-life $t_{1/2}$ of the sample is 10 hours. The initial velocity v_0 is 9 m/s and the final velocity v_f is 16 m/s. The mass of the sample is 1 times 10^{-3} kg.

Describing Data

Data are described by a centre and a spread. Small symbols attached to letters keep the many quantities apart.

The square of the standard deviation, s^2, is called the variance. The roots of the quadratic are $x_1 = 3$ and $x_2 = 5$. The point (x_1, y_1) lies on the line, and so does (x_2, y_2). A tolerance of 10^{-3} is enough for the 19th iteration of the method.

Expanding $(a + b)^2$ gives $a^2 + 2ab + b^2$. The coefficient of determination R^2 equals 0.84 for this fit. Each term satisfies $a_{n+1} = a_n + 2$, so with $a_1 = 3$ the 5th term is 11. The initial velocity v_0 is 7 m/s and the final velocity v_f is 10 m/s.

The square of the standard deviation, s^2, is called the variance. The roots of the quadratic are $x_1 = 2$ and $x_2 = 7$. The mass of the sample is 2.5 times 10^{-3} kg. The null hypothesis H_0 is rejected when the p-value is below 0.05.

How do we describe the spread of data?

Example

The sum of two numbers is 25 and their difference is 7. Find both numbers.

Solution

Adding the two equations gives twice the larger number, 32, so the larger number is 16 and the smaller is 9.

Key Takeaways

- Round only at the end of a calculation.
- Keep two spare digits in every intermediate step.
- Report the answer with the precision of the data.

Table 1: Measured densities of four materials

Material	Mass (g)	Volume (cm³)	Density (g/cm³)
Aluminium	67.5	25	2.70
Iron	196.8	25	7.87
Copper	448.0	50	8.96
Ice	46.0	50	0.92
Granite	137.5	50	2.75

Unit 5: Practice and Review

Practice is the quickest way to find out what has been understood. The following exercises use only the notation of the units above.

Practice Exercises

Solve each equation, and give a reason for every step.

1. $2/x = 5/5$
2. $2x + 5 = 9 \times 3$. Check by substitution.
3. $\sqrt{(x + 7)} = 11$
4. $|x - 13| \le 5$
5. $f(x) = 2x^3 - 1x$
6. $5x - 3y = 18$. Find all solutions.
7. $0 = 9x^2 - 15x + 1$
8. $(x - 9)^2 + (y + 1)^2 = 144$. Explain your steps.
9. $y = (3x + 5) / (x - 15)$. Solve for x.
10. $\sin \theta = 11/13$
11. $5x^2 + 1x = 0$. Explain your steps.
12. $x/3 + y/13 = 1$
13. $2x + 13 \ge 9x - 7$
14. $x^2 - 49 = 0$
15. $|x - 7| \le 11$. Explain your steps.
16. $5x + 9 \ge 3x - 7$. Check by substitution.
17. $3/x = 3/5$. Find all solutions.
18. $2x + 7 = 5 \times 4$. Solve for x.
19. $2x^2 + 15x = 0$
20. $y = (3x + 1) / (x - 3)$
21. $\sin \theta = 11/16$
22. $7x - 13y = 18$. Solve for x.
23. $\sqrt{(x + 3)} = 5$
24. $f(x) = 5x^3 - 5x$. Solve for x.
25. $0 = 9x^2 - 13x + 9$. Find all solutions.
26. $x^2 - 25 = 0$
27. $(x - 7)^2 + (y + 15)^2 = 9$
28. $x/7 + y/7 = 1$. Solve for x.

Key Terms

A good set of notes ends with the terms that were used. The table lists them in alphabetical order with a short definition.

Term	Definition
Derivative	The limit of the ratio of the change in a function to the change in its input; it gives the slope of the tangent line.

Term	Definition
Integral	A limit of sums that measures accumulated change, such as the area under a curve.
Limit	The value that a function approaches as its input gets arbitrarily close to a chosen point.
Matrix	A rectangular array of numbers arranged in rows and columns.
Determinant	A single number computed from a square matrix that tells whether the matrix can be inverted.
Eigenvalue	A scalar by which a matrix stretches a special direction called an eigenvector.
Vector	A quantity with both a size and a direction, often written as a column of numbers.
Scalar	A quantity described by a single number and, in physics, a unit.
Domain	The set of all inputs for which a function is defined.
Range	The set of all outputs that a function actually produces.

Reading and Schedule

Reading list. Chapter 3 of the notes and the short article on measurement error.

Schedule. Lecture on Monday, laboratory on Wednesday and a quiz on Friday.